

The Lovász number is not alone:

a continuum of Rényi points in the entanglement-assisted spectrum of graphs

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Abstract

Li and Zuiddam’s entanglement-assisted asymptotic spectrum of graphs collects the graph parameters that behave like capacities for zero-error communication with shared entanglement. To our knowledge, its only previously known element was the Lovász number ϑ . We find a continuum of others.

The starting point is an observation about three quantities from Duan and Winter’s work on no-signalling-assisted zero-error communication. The Lovász number, the smallest Holevo capacity $C_{\min}$ and the no-signalling simulation cost Σ of a graph are the same kind of object: information radii of the classical–quantum channels compatible with the graph, measured with Petz–Rényi divergences of orders 0, 1 and 2. Interpolating in the order gives a family ϑ_α , $0 \leq \alpha \leq 2$, and we prove that every member lies in the spectrum. The main step is multiplicativity under the strong product. It has to hold for representations of the product that do not factor, and an exact nesting identity for power compressions takes care of this.

On the five-cycle the family sweeps out the whole interval $[\sqrt{5}, 5/2]$, so it contains continuum many distinct points. A single Hilbert–Schmidt inequality gives $C_{\min}(C_5) = \log_2(5/2)$, which proves the pentagon conjecture of Duan and Winter. The same framework settles the two links of their inequality chain that they left open. The asymptotic no-signalling simulation cost of a graph always equals its one-shot value $\log_2 \Sigma$. And a 32-vertex graph built from the Mermin–Peres magic square, certified in exact rational arithmetic, has $C_{\min} < \log_2 \Sigma$.

1 Introduction

Shannon’s zero-error capacity $\Theta(G)$ of a graph is notoriously hard to compute [Sha56]. Zuiddam found a clean way to organize what we know about it [Zui19]. Consider every graph parameter that adds under disjoint union, multiplies under the strong product, equals 1 on a single vertex and respects the cohomomorphism preorder. These parameters form the *asymptotic spectrum of graphs*, and Strassen’s duality theorem [Str88] says that Θ is their pointwise minimum. So every spectral point is an upper bound on the Shannon capacity, and many of the classical upper bounds turn out to be spectral points: the Lovász number, the fractional Haemers bounds, the complement of the projective rank, and the fractional clique cover number. The ordinary spectrum is by now known to be uncountable [Vra21].

Li and Zuiddam carried the theory over to quantum and entanglement-assisted communication [LZ21]. With shared entanglement the preorder changes. Now $G \leq_* H$ means that, with the help of an entangled state, one use of a channel with confusability graph H can do the job of one use of a channel with confusability graph G . The duality survives: the entanglement-assisted Shannon capacity Θ_* is the minimum over a new spectrum $X(\mathcal{G}, \leq_*)$. This spectrum has proved much harder to populate. Beigi had shown that $\Theta_* \leq \vartheta$ [Bei10], and Li and Zuiddam proved that ϑ

itself lies in $X(\mathcal{G}, \leq_*)$. They found no other element. They also suggested that the conjecture $\Theta_* = \vartheta$ might be proved by showing that ϑ is the least element of the spectrum, “or even the only point” [LZ21].

It is not the only point. This paper constructs a continuum of others.

1.1 Three quantities, one family

We start from a paper of Duan and Winter [DW16] on zero-error communication assisted by no-signalling correlations. For a confusability graph G they compare several numbers attached to the classical–quantum channels $x \mapsto \rho_x$ whose confusability graph is G . There is the Lovász number $\vartheta(G)$. There is $C_{\min}(G)$, the smallest Holevo capacity of such a channel, and $\Sigma(G)$, the smallest cost of simulating such a channel exactly with no-signalling help. They prove the chain

$$\log_2 \vartheta(G) \leq C_{\min}(G) \leq S_{0,\text{NS}}(G) \leq \log_2 \Sigma(G) \leq \log_2 \alpha^*(G), \quad (1)$$

where $S_{0,\text{NS}}$ is an asymptotic simulation cost and α^* is the fractional packing number (Section 2.2 has the definitions).

The first three quantities in (1) are really one quantity in disguise. Fix a channel compatible with G and a state σ . Measure the distance from each output ρ_x to σ with the Petz–Rényi divergence D_α and take the worst case. This worst case, $\max_x D_\alpha(\rho_x \parallel \sigma)$, is an order- α *information radius*. Minimizing it over channels and centers gives a graph parameter:

$$\log_2 \vartheta_\alpha(G) = \inf_{(\rho_x), \sigma} \max_{x \in V(G)} D_\alpha(\rho_x \parallel \sigma), \quad 0 \leq \alpha \leq 2. \quad (2)$$

Then

$$\vartheta_0 = \vartheta, \quad \vartheta_1 = 2^{C_{\min}}, \quad \vartheta_2 = \Sigma. \quad (3)$$

The middle identity is the familiar min–max formula for the Holevo capacity. The other two are elementary once the right formula is in place (Theorem 3.3). So the three Duan–Winter quantities are samples of one curve $\alpha \mapsto \vartheta_\alpha(G)$, taken at orders 0, 1 and 2. This also explains why they come in the order they do, since D_α grows with α .

What happens in between?

1.2 Results

Theorem A (every order up to two is a spectral point). *For every $\alpha \in [0, 2]$, the graph parameter ϑ_α belongs to the entanglement-assisted asymptotic spectrum $X(\mathcal{G}, \leq_*)$.*

Normalization and additivity are routine. Monotonicity under $\leq_*$ is a data-processing argument, and it is why the order stops at two: that is where Petz divergences stop satisfying data processing. The real work is multiplicativity, $\vartheta_\alpha(G \boxtimes H) = \vartheta_\alpha(G) \vartheta_\alpha(H)$. One inequality follows by tensoring witnesses. The other has to control representations of $G \boxtimes H$ that are *not* tensor products, and there is no reason to expect an optimal representation of a product to factor. Section 4 handles this with an exact nesting identity for the compressions that appear in (2).

Are the new points actually new? The pentagon answers this.

Theorem B (the pentagon). *For $\frac{1}{2} \leq \alpha \leq 2$ we have $\vartheta_\alpha(C_5) = \frac{5}{2}$. For $0 \leq \alpha \leq \frac{1}{2}$, writing $p = 1/(1 - \alpha) \in [1, 2]$,*

$$\sqrt{5} \left(\frac{\sqrt{5}}{2} \right)^{p-1} \leq \vartheta_\alpha(C_5) \leq \min \left\{ \frac{5}{2}, \sqrt{5} \cdot 5^{(p-1)/2} \right\}. \quad (4)$$

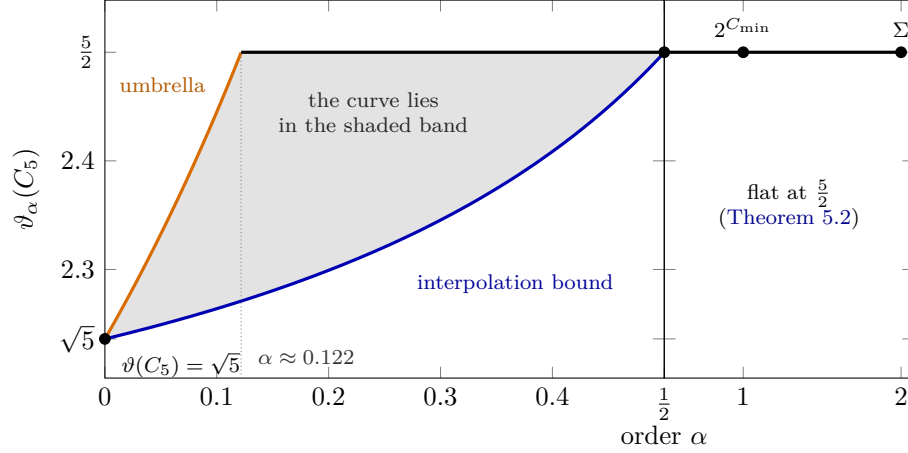


Figure 1: What is known about $\alpha \mapsto \vartheta_\alpha(C_5)$. The horizontal scale changes at $\alpha = \frac{1}{2}$. On $[\frac{1}{2}, 2]$ the curve is exactly $\frac{5}{2}$, which gives $C_{\min}(C_5) = \log_2 \frac{5}{2}$ and $\Sigma(C_5) = \frac{5}{2}$. On $[0, \frac{1}{2}]$ it is continuous and nondecreasing and lies between the interpolation lower bound (blue) and the minimum of $\frac{5}{2}$ and the Lovász-umbrella bound (orange). The umbrella bound is better than $\frac{5}{2}$ only for $\alpha < 1 - \ln 5 / (2 \ln \frac{5}{2}) \approx 0.122$. Continuity and the endpoint values $\sqrt{5}$ and $\frac{5}{2}$ force the curve through every value in between.

The map $\alpha \mapsto \vartheta_\alpha(C_5)$ is continuous and carries $[0, \frac{1}{2}]$ onto $[\sqrt{5}, \frac{5}{2}]$. In particular $X(\mathcal{G}, \leq_*)$ contains continuum many distinct points of the form ϑ_α .

Figure 1 shows what we know about the pentagon curve. Since $\vartheta_1(C_5) = \vartheta_2(C_5) = 5/2$, Theorem B gives the values of $C_{\min}$ and Σ on the pentagon. That is precisely what Duan and Winter conjectured right after (1), where they note that a rigorous proof “has so far eluded” them [DW16, Eq. (54) and the paragraph after it].

Corollary (the Duan–Winter pentagon conjecture). $\log_2 \sqrt{5} < C_{\min}(C_5) = S_{0, \text{NS}}(C_5) = \log_2 \Sigma(C_5) = \log_2 \alpha^*(C_5) = \log_2 \frac{5}{2}$.

The pentagon cannot tell order one from order two. A different graph can.

Theorem C (order one is not order two). *There is a graph G_{32} on 32 vertices, built from the Mermin–Peres magic square, such that*

$$2^{C_{\min}(G_{32})} \leq 6.186983\dots < 6.194232\dots \leq \Sigma(G_{32}).$$

Both outer bounds are certified by exact rational data. In particular $\vartheta_1 \neq \vartheta_2$.

1.3 A taste of the method: the pentagon in one inequality

The pentagon conjecture has a proof short enough to give here in full. Label C_5 by $\mathbb{Z}_5$ with $i \sim i \pm 1$. In a representation of the pentagon, non-neighbours get orthogonal projections: $P_i P_{i+2} = 0$ for all i . At order $\alpha = \frac{1}{2}$, the formula of Section 3 for ϑ_α becomes

$$\frac{1}{\vartheta_{1/2}(G)} = \sup_{(P_x), \sigma} \min_x \|P_x \sigma^{1/2} P_x\|_2^2,$$

the supremum running over representations (P_x) of G and faithful states σ . Now the one inequality we need: for every representation of C_5 and every matrix X ,

$$\sum_{i \in \mathbb{Z}_5} \|P_i X P_i\|_2^2 \leq 2 \|X\|_2^2. \quad (5)$$

The reason is local. The two neighbours $i - 1$ and $i + 1$ of a vertex are not adjacent to each other, so $P_{i-1} + P_{i+1}$ is again a projection. Seen from inside P_i , the sum of all five projections is therefore at most twice the identity ([Theorem 5.1](#)). Put $X = \sigma^{1/2}$. The right side of (5) is $2 \operatorname{Tr} \sigma = 2$, so some vertex has $\|P_i \sigma^{1/2} P_i\|_2^2 \leq 2/5$, and $\vartheta_{1/2}(C_5) \geq 5/2$. Since ϑ_α increases with α , the same bound holds for $\vartheta_1 = 2^{C_{\min}}$ and for $\vartheta_2 = \Sigma$.

The matching upper bound is the obvious classical channel: input i produces a uniformly random element of $\{i, i + 1\}$. Its Holevo capacity is $\log_2 5 - 1 = \log_2 \frac{5}{2}$, and $T = \frac{1}{2}I$ dominates every output with $\operatorname{Tr} T = \frac{5}{2}$. Duan and Winter’s chain (1) squeezes $S_{0,\text{NS}}$ in between, and $\alpha^*(C_5) = \frac{5}{2}$ is classical. That is the whole proof.

1.4 What is new

New points in the entanglement-assisted spectrum. To our knowledge, the parameters ϑ_α with $\alpha > 0$ are the first elements of $X(\mathcal{G}, \leq_*)$ other than ϑ . [Theorem B](#) settles Li and Zuiddam’s “only point” question in the negative. It says nothing about whether ϑ is the *least* point. Every ϑ_α dominates ϑ , so the bound $\Theta_* \leq \vartheta$ is not improved and the conjecture $\Theta_* = \vartheta$ is untouched.

Multiplicativity for joint representations. Duan and Winter proved that Σ is multiplicative for two *fixed* cq-graphs [[DW16](#)]. In our language, that is the case of [Theorem 4.1](#) in which the representation of $G \boxtimes H$ is a tensor product. [Theorem 4.1](#) drops that assumption, at every order in $[0, 2]$. At the two lower endpoints this recovers known results: order zero is Lovász’s multiplicativity of ϑ [[Lov79](#)], and order one is Duan and Winter’s additivity of $C_{\min}$ under the strong product [[DW16](#), Lemma 23]. Every order in $(0, 1) \cup (1, 2]$ is new, and order two is the one with an operational consequence. There it gives a single-letter formula,

$$S_{0,\text{NS}}(G) = \log_2 \Sigma(G),$$

for the cheapest asymptotic no-signalling simulation of a channel with confusability graph G . Duan and Winter single out asymptotic simulation with no-signalling assistance as an important open problem. At the level of confusability graphs, this reduces it to a one-shot problem.

The Duan–Winter chain, link by link. With (3) in hand, every link of (1) can now be decided ([Fig. 2](#)). Duan and Winter showed that the last link can be strict. They conjectured that the first is strict on the pentagon, which is the corollary above. They left open whether either of the two middle links can be strict. The third link is always an equality ([Theorem A](#) at order two). The second can be strict ([Theorem C](#)).

One curve instead of three numbers. Duan and Winter already wrote $\log_2 \Sigma$ as a conditional min-entropy, which is a max-relative-entropy radius. What seems to be new is that, once one minimizes over states supported in a fixed projection, this radius coincides with the *Petz* radius of order two. All three quantities then sit on a single monotone curve. Above order one the curve is also the same for the sandwiched Rényi divergences: the sandwiched radius of order $\beta \geq 1$ equals

order $\alpha = 0$		order $\alpha = 1$		asymptotic		order $\alpha = 2$		LP bound
$\log_2 \vartheta$	$\leq$	$C_{\min}$	$\leq$	$S_{0,\text{NS}}$	$\leq$	$\log_2 \Sigma$	$\leq$	$\log_2 \alpha^*$
		strict on C_5 conjectured by DW; proved here, Thm B		can be strict open in DW; G_{32} , Thm C		always equal open in DW; Thm A at $\alpha = 2$		can be strict shown by DW
C_5 :	1.161		1.322		1.322		1.322	1.322
G_{32} :	.	≤ 2.6293		≥ 2.6309		≥ 2.6309		.

Figure 2: The Duan–Winter chain (1) after this paper. The three quantities in the first four columns that carry an order are the points $\alpha = 0, 1, 2$ of one curve. Shaded boxes record the status of each link; teal marks what is new here. The bottom rows give values in bits (dots are values we do not need).

$\vartheta_{2-1/\beta}$ (Theorem 3.5), so nothing depends on which quantum Rényi divergence one prefers there. The picture has a useful by-product: every ϑ_α is squeezed between ϑ and α^* by flat witnesses (Theorem 3.3). So the family collapses to a single value on graphs with $\vartheta = \alpha^*$, perfect graphs among them, and it can only separate points on graphs with $\vartheta < \alpha^*$.

1.5 Related work

Dalai showed that the Lovász bound and the sphere-packing bound for classical–quantum channels are both information radii of Csiszár type built from Rényi divergences [Dal13]. Our order-zero endpoint is an object of the same kind, and the family ϑ_α continues it to positive order at the level of graphs. As explained above, Duan and Winter’s fixed-cq-graph multiplicativity is the factoring case of our product theorem. In the ordinary spectrum, continua arise in a different way. Vrana’s probabilistic refinement makes the ordinary spectrum log-convex in a suitable sense [Vra21], and so interpolating between two known points produces new ones. We do not know whether an analogue holds for $\leq_*$. Section 7 explains why, even if it does, at least ϑ_1 is not such an interpolation. For the weaker quantum and commuting-quantum preorders, Haemers-type parameters give further points [LZ21, GGL22]. Their monotonicity under $\leq_*$ is not known. Finally, Vrana studied which points of the ordinary spectrum extend to non-commutative graphs [Vra22]. We stay with classical graphs throughout.

Organization. Section 2 fixes conventions. Section 3 rewrites ϑ_α in terms of projections and proves the basic properties of the family. Section 4 proves Theorem A. Section 5 treats the pentagon and the Duan–Winter chain, and Section 6 separates orders one and two. Open problems are collected in Section 7. The appendices contain the certificate format and a few routine proofs.

2 Setup

Conventions

- Graphs are finite and simple, and they are read as *confusability graphs*: $x \sim y$ means that inputs x and y may be confused. We write $x \simeq y$ for “ $x = y$ or $x \sim y$ ”.
- Distinct non-adjacent vertices must be told apart, so they are represented by *orthogonal* objects. This is Lovász’s original orientation [Lov79]. Li and Zuiddam express the same relations through graph complements.
- $G \sqcup H$ is the disjoint union and $G \boxtimes H$ the strong product, where $(x, y) \simeq (x', y')$ if and only if $x \simeq x'$ and $y \simeq y'$. K_1 is a single vertex, $\overline{K_n}$ is the edgeless graph on n vertices (a noiseless channel with n inputs), and C_5 is the pentagon.
- Logarithms are to base two unless written $\ln$. Hilbert spaces are finite-dimensional, of arbitrary dimension. A state is *faithful* if it is invertible. Π_ρ is the support projection of ρ .

2.1 Representations and Rényi radii

Definition 2.1. A *representation* of G is a family $(P_x)_{x \in V(G)}$ of nonzero orthogonal projections on a Hilbert space such that $P_x P_y = 0$ whenever $x \neq y$ are non-adjacent. A family of states (ρ_x) is *compatible with G* if its supports form a representation.

A compatible family is the same thing as a classical–quantum channel $x \mapsto \rho_x$ whose confusability graph is contained in G : two inputs can only be confused if they are adjacent. Extra orthogonality between adjacent vertices is allowed. Duan and Winter optimize over the same class of channels. Readers who prefer channels whose confusability graph is exactly G lose nothing, because every infimum below is the same for both classes (Theorem B.1).

For a state ρ and a faithful state σ , the Petz–Rényi divergence of order $\alpha \in (0, 1) \cup (1, 2]$ is

$$D_\alpha(\rho \parallel \sigma) = \frac{1}{\alpha - 1} \log \text{Tr } \rho^\alpha \sigma^{1-\alpha}.$$

At $\alpha = 1$ it is the relative entropy $D(\rho \parallel \sigma)$, and at $\alpha = 0$ it is $D_0(\rho \parallel \sigma) = -\log \text{Tr } \Pi_\rho \sigma$. We use four standard facts [Pet86, Tom16]. The divergence is nonnegative and vanishes at $\rho = \sigma$. It is additive under tensor products. It is nondecreasing in α . And it satisfies data processing for every $\alpha \in [0, 2]$.

Definition 2.2. For a graph G and $\alpha \in [0, 2]$, define $\vartheta_\alpha(G)$ by

$$\log \vartheta_\alpha(G) = \inf \left\{ \max_x D_\alpha(\rho_x \parallel \sigma) : (\rho_x) \text{ compatible with } G, \sigma \text{ faithful} \right\}, \quad (6)$$

where the states live on a common Hilbert space of any finite dimension. For the empty graph we set $\vartheta_\alpha(K_0) = 0$.

Nothing changes if singular centers are allowed with the usual support conventions: mix a singular center with a little of the maximally mixed state and let the weight go to zero.

2.2 The Duan–Winter quantities

Duan and Winter attach the following quantities to a graph [DW16]. The *minimal Holevo capacity* is

$$C_{\min}(G) = \inf_{(\rho_x) \text{ compatible}} \max_p \left[S \left(\sum_x p_x \rho_x \right) - \sum_x p_x S(\rho_x) \right]. \quad (7)$$

The *no-signalling simulation cost* is

$$\Sigma(G) = \inf\{\text{Tr } T : T \geq \rho_x \text{ for all } x, (\rho_x) \text{ compatible}\}. \quad (8)$$

Duan and Winter show that $\min\{\text{Tr } T : T \geq \rho_x \forall x\}$ governs the cost of simulating the channel $x \mapsto \rho_x$ exactly with no-signalling assistance. So $\log \Sigma(G)$ is the cheapest such simulation among channels compatible with G . Tensoring witnesses shows that Σ is submultiplicative under $\boxtimes$, so the regularization

$$S_{0,\text{NS}}(G) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \Sigma(G^{\boxtimes n})$$

exists. This is Duan and Winter's graph-level asymptotic simulation cost. Finally, $\alpha^*(G)$ is the fractional packing number: the largest $\sum_x v_x$ over $v \geq 0$ with $\sum_{x \in C} v_x \leq 1$ for every clique C . By LP duality it is also the smallest $\sum_C w_C$ over $w \geq 0$ with $\sum_{C \ni x} w_C \geq 1$ for every vertex x .

2.3 The entanglement-assisted spectrum

We use the entanglement-assisted cohomomorphism preorder of Li and Zuiddam [LZ21], which comes from the entanglement-assisted homomorphisms of Cubitt et al. [CMR⁺14]. In confusability language, $G \leq_* H$ means that there are a state ω and positive semidefinite operators ω_g^h ($g \in V(G)$, $h \in V(H)$) with

$$\sum_{h \in V(H)} \omega_g^h = \omega \quad \text{for every } g, \quad (9)$$

$$\omega_g^h \omega_{g'}^{h'} = 0 \quad \text{whenever } g \neq g' \text{ are non-adjacent in } G \text{ and } h \simeq h' \text{ in } H. \quad (10)$$

Here is the operational picture. Alice and Bob share a pure entangled state, and ω is Bob's half of it. To send g , Alice measures her half with a measurement that depends on g , obtains an outcome h , and sends h through a channel with confusability graph H . Her measurement steers Bob's half to ω_g^h (unnormalized), and (9) says that Bob's marginal does not depend on g . Suppose g and g' must be told apart but the channel might confuse h with h' . Then (10) lets Bob tell them apart anyway, by measuring his own half. So $G \leq_* H$ says that, with entanglement, one use of an H -channel does everything one use of a G -channel does. For example, $\overline{K_n} \leq_* G$ says that n messages can be sent without error over one use of G .

Definition 2.3. The *entanglement-assisted asymptotic spectrum* $X(\mathcal{G}, \leq_*)$ is the set of maps ϕ from finite graphs to $[0, \infty)$ such that

$$\phi(G \sqcup H) = \phi(G) + \phi(H), \quad \phi(G \boxtimes H) = \phi(G)\phi(H), \quad \phi(K_1) = 1, \quad G \leq_* H \Rightarrow \phi(G) \leq \phi(H).$$

Li and Zuiddam show that $\Theta_*(G) = \min_{\phi \in X(\mathcal{G}, \leq_*)} \phi(G)$, and that $\vartheta \in X(\mathcal{G}, \leq_*)$ [LZ21].

3 Power compressions and the shape of the family

The definition (6) optimizes over states. It is much easier to work with projections, and the tool for this is the following compression.

3.1 From states to projections

Let $P \neq 0$ be a projection, let σ be faithful, and let $s \in [-1, 1]$. On $\text{ran } P$ define

$$K_s(\sigma, P) = \begin{cases} (P\sigma^s P)^{1/s}, & s \neq 0, \\ \exp(P(\ln \sigma)P), & s = 0, \end{cases} \quad Z_s(\sigma, P) = \text{Tr } K_s(\sigma, P),$$

extended by zero off $\text{ran } P$. We call K_s the *power compression* of σ to P and Z_s its *weight*. Four values of s recur. $Z_1 = \text{Tr } P\sigma$ is the plain weight. $Z_{1/2} = \|P\sigma^{1/2}P\|_2^2$ is a Hilbert–Schmidt weight. Z_0 is a Gibbs weight. And $Z_{-1} = \text{Tr}(P\sigma^{-1}P)^{-1}$ is the trace of the *shorted operator* $\sigma_{/P}$, the largest positive operator below σ with range in $\text{ran } P$ [AT75]. All the weights are homogeneous of degree one: $K_s(c\sigma, P) = c K_s(\sigma, P)$ for $c > 0$.

Proposition 3.1 (supported minimization). *Let $0 \leq \alpha \leq 2$ and $s = 1 - \alpha$. Then*

$$\min_{\rho: \text{supp } \rho \subseteq \text{ran } P} D_\alpha(\rho \| \sigma) = -\log Z_s(\sigma, P),$$

and the minimum is attained at $\rho = K_s(\sigma, P)/Z_s(\sigma, P)$.

Proof. For ρ supported in P we have $\text{Tr } \rho^\alpha \sigma^s = \text{Tr } \rho^\alpha A$ with $A = P\sigma^s P$, and $Z_s = \text{Tr } A^{1/s}$ when $s \neq 0$.

Case $0 < \alpha < 1$. Here $s \in (0, 1)$. Hölder’s inequality for Schatten norms gives $\text{Tr } \rho^\alpha A \leq \|\rho^\alpha\|_{1/\alpha} \|A\|_{1/s} = (\text{Tr } A^{1/s})^s$. Since $\alpha - 1 < 0$, this becomes $D_\alpha(\rho \| \sigma) \geq \frac{s}{\alpha-1} \log Z_s = -\log Z_s$, with equality at $\rho \propto A^{1/s}$.

Case $1 < \alpha \leq 2$. Now $s \in [-1, 0)$ and A is positive definite on $\text{ran } P$. Write $A = \sum_k a_k |e_k\rangle\langle e_k|$ and $q_k = \langle e_k | \rho | e_k \rangle$, a probability vector. Convexity of $t \mapsto t^\alpha$ gives $\langle e_k | \rho^\alpha | e_k \rangle \geq q_k^\alpha$, so $\text{Tr } \rho^\alpha A \geq \sum_k a_k q_k^\alpha$. By Hölder,

$$1 = \sum_k (a_k^{1/\alpha} q_k) a_k^{-1/\alpha} \leq \left(\sum_k a_k q_k^\alpha \right)^{1/\alpha} \left(\sum_k a_k^{1/s} \right)^{(\alpha-1)/\alpha},$$

using $-1/(\alpha-1) = 1/s$. Hence $\text{Tr } \rho^\alpha A \geq (\sum_k a_k^{1/s})^s = Z_s^s$, and dividing the logarithm by $\alpha-1 > 0$ gives $D_\alpha \geq -\log Z_s$. Equality holds when ρ is diagonal in (e_k) with $q_k \propto a_k^{1/s}$, that is, at $\rho \propto A^{1/s}$.

Case $\alpha = 1$. On $\text{ran } P$ we have $\ln K_0 = P(\ln \sigma)P$, so $\text{Tr } \rho \ln \sigma = \text{Tr } \rho \ln(K_0/Z_0) + \ln Z_0$ and $D(\rho \| \sigma) = D(\rho \| K_0/Z_0) - \log Z_0 \geq -\log Z_0$.

Case $\alpha = 0$. $\text{Tr } \Pi_\rho \sigma \leq \text{Tr } P\sigma = Z_1$, with equality whenever ρ has full support in $\text{ran } P$. $\square$

The minimizers for different vertices have the supports we prescribe, so they form a compatible family whenever the projections form a representation. Conversely, the supports of any compatible family form a representation. This proves the projection formula:

$$\frac{1}{\vartheta_\alpha(G)} = \sup_{(P_x), \sigma} \min_{x \in V(G)} Z_{1-\alpha}(\sigma, P_x), \quad (11)$$

with the supremum over representations of G and faithful states σ on the same space.

3.2 Endpoints, flat witnesses and a sandwich

Some witnesses have the same radius at every order. Call a pair (ρ, σ) *flat* if σ commutes with a projection Π and $\rho = \Pi\sigma / \text{Tr } \Pi\sigma$.

Lemma 3.2 (flat witnesses). *If (ρ, σ) is flat, then $D_\alpha(\rho||\sigma) = -\log \text{Tr } \Pi\sigma$ for every $\alpha \in [0, 2]$.*

Proof. With $c = \text{Tr } \Pi\sigma$ we get $\text{Tr } \rho^\alpha \sigma^{1-\alpha} = c^{-\alpha} \text{Tr } \Pi\sigma = c^{1-\alpha}$, which gives the claim for $\alpha \neq 0, 1$. The two remaining orders are direct. $\square$

Proposition 3.3. *For every graph G :*

- (i) $\vartheta_0(G) = \vartheta(G)$, $\vartheta_1(G) = 2^{C_{\min}(G)}$ and $\vartheta_2(G) = \Sigma(G)$;
- (ii) $\alpha \mapsto \vartheta_\alpha(G)$ is nondecreasing on $[0, 2]$;
- (iii) $\vartheta(G) \leq \vartheta_\alpha(G) \leq \alpha^*(G)$ for every $\alpha \in [0, 2]$.

Proof. (i) *Order zero.* By (11), $1/\vartheta_0(G) = \sup \min_x \text{Tr } P_x \sigma$. Lovász's handle form reads $1/\vartheta(G) = \sup \min_x |\langle c, u_x \rangle|^2$ over orthonormal representations (u_x) of G and unit vectors c [Lov79]. A vector representation is a projection representation with $P_x = |u_x\rangle\langle u_x|$, and a faithful approximation of $|c\rangle\langle c|$ gives $\vartheta_0 \leq \vartheta$. Conversely, purify σ to a unit vector $c \in \mathcal{H} \otimes \mathcal{H}'$ and put $u_x \propto (P_x \otimes I)c$. Non-adjacent vertices get orthogonal vectors, and $|\langle c, u_x \rangle|^2 = \text{Tr } P_x \sigma$, so $\vartheta \leq \vartheta_0$.

Order one. For each compatible family, the Holevo capacity equals the relative-entropy radius $\min_\sigma \max_x D(\rho_x||\sigma)$ [SW01]. Taking the infimum over families gives $\vartheta_1 = 2^{C_{\min}}$.

Order two. By (11), $1/\vartheta_2 = \sup \min_x \text{Tr } \sigma_{/P_x}$. Given a compatible family and $T \geq \rho_x$ with $t = \text{Tr } T$, put $\sigma = T/t$, regularized if needed. Each ρ_x/t lies below σ with range in $\text{ran } P_x$, so $\text{Tr } \sigma_{/P_x} \geq 1/t$. Hence $\vartheta_2 \leq \Sigma$. Conversely, if $\text{Tr } \sigma_{/P_x} \geq c$ for all x , the states $\rho_x = \sigma_{/P_x} / \text{Tr } \sigma_{/P_x}$ are compatible and all lie below $T = \sigma/c$, whose trace is $1/c$. Hence $\Sigma \leq \vartheta_2$.

(ii) D_α is nondecreasing in α for each pair of states, and infima preserve this.

(iii) The lower bound follows from (i) and (ii). For the upper bound take a fractional clique cover w with $\sum_C w_C = \alpha^*(G)$, and work on the space spanned by orthonormal vectors $|C\rangle$, one for each clique with $w_C > 0$. Put $\sigma = \sum_C w_C |C\rangle\langle C| / \alpha^*(G)$ and let Π_x project onto the cliques containing x . Non-adjacent vertices share no clique, so the flat states $\rho_x = \Pi_x \sigma / \text{Tr } \Pi_x \sigma$ are compatible. By Theorem 3.2 their radius at every order is $\max_x \log(\alpha^*(G) / \sum_{C \ni x} w_C) \leq \log \alpha^*(G)$. $\square$

Two consequences are worth stating. First, if $\vartheta(G) = \alpha^*(G)$, the whole family collapses on G . This happens for every perfect graph. Second, if some flat witness is optimal for $\vartheta_1(G)$, then $\vartheta_1(G) = \vartheta_2(G)$, because the flat radius does not depend on the order. On the pentagon, this is exactly what happens (Section 5).

3.3 Continuity

Proposition 3.4. *For every graph G , the map $\alpha \mapsto \vartheta_\alpha(G)$ is continuous on $[0, 1]$.*

Right continuity comes from fixing a near-optimal witness. Left continuity comes from a comparison $D_\alpha \leq \frac{\alpha(1-\beta)}{\beta(1-\alpha)} D_\beta$ for $\beta < \alpha < 1$, whose constant does not depend on the states or the dimension. The details are in Section B.1.

3.4 Sandwiched radii give the same curve

The Petz divergence is not the only quantum Rényi divergence. For $\beta \in (1, \infty)$ the *sandwiched* Rényi divergence [MLDS⁺13, WWY14] is

$$\tilde{D}_\beta(\rho\|\sigma) = \frac{1}{\beta-1} \log \text{Tr} \left(\sigma^{\frac{1-\beta}{2\beta}} \rho \sigma^{\frac{1-\beta}{2\beta}} \right)^\beta,$$

with the relative entropy $\tilde{D}_1 = D$ and the max-relative entropy $\tilde{D}_\infty(\rho\|\sigma) = \log \min\{\lambda : \rho \leq \lambda\sigma\}$ at the two ends. Define $\tilde{\vartheta}_\beta(G)$ by (6) with $\tilde{D}_\beta$ in place of D_α . Above order one, this gives nothing new.

Proposition 3.5 (sandwiched radii). *For every graph G and every $\beta \in [1, \infty]$,*

$$\tilde{\vartheta}_\beta(G) = \vartheta_{2-1/\beta}(G),$$

with $2 - 1/\infty = 2$. In particular $\tilde{\vartheta}_1 = 2^{C_{\min}}$ and $\tilde{\vartheta}_\infty = \Sigma$.

Proof. Put $\theta = (\beta - 1)/\beta \in [0, 1]$, so that $s = -\theta$ is the parameter of the Petz order $\alpha = 1 - s = 2 - 1/\beta$. The argument leading to (11) applies verbatim once we know the supported minimum: for a projection $P \neq 0$ and a faithful state σ ,

$$\min_{\rho: \text{supp } \rho \subseteq \text{ran } P} \tilde{D}_\beta(\rho\|\sigma) = -\log Z_{-\theta}(\sigma, P). \quad (12)$$

At $\beta = 1$ this is Theorem 3.1 at order one. At $\beta = \infty$, the condition $\rho \leq \lambda\sigma$ for a state supported in $\text{ran } P$ says that ρ/λ lies below σ with range in $\text{ran } P$, so $1/\lambda \leq \text{Tr } \sigma|_P = Z_{-1}(\sigma, P)$, with equality at $\rho = \sigma|_P / \text{Tr } \sigma|_P$.

Now let $1 < \beta < \infty$. Put $A = P\sigma^{-\theta}P$, which is positive definite on $\text{ran } P$, and $X = A^{1/2}\rho A^{1/2}$. The operator $\sigma^{-\theta/2}\rho\sigma^{-\theta/2}$ has the same nonzero eigenvalues as $\rho^{1/2}\sigma^{-\theta}\rho^{1/2} = \rho^{1/2}A\rho^{1/2}$, and hence as X , so the sandwiched moment is $\text{Tr } X^\beta$. Since $\text{Tr } A^{-1}X = \text{Tr } \rho = 1$, Hölder's inequality with the conjugate exponents β and $\beta/(\beta - 1) = 1/\theta$ gives

$$1 \leq \|X\|_\beta \|A^{-1}\|_{1/\theta} = (\text{Tr } X^\beta)^{1/\beta} (\text{Tr } A^{-1/\theta})^\theta.$$

Here $A^{-1/\theta} = (P\sigma^{-\theta}P)^{1/(-\theta)} = K_{-\theta}(\sigma, P)$, so $\text{Tr } X^\beta \geq Z_{-\theta}(\sigma, P)^{-(\beta-1)}$, which is the lower bound in (12). At $\rho = K_{-\theta}/Z_{-\theta}$ we get $X \propto A^{1-1/\theta}$, so $X^\beta \propto A^{-1/\theta} = (A^{-1})^{1/\theta}$, which is the equality case of Hölder's inequality. $\square$

So above order one the curve does not depend on which of the two standard quantum Rényi divergences is used: the sandwiched radii of orders $\beta \in [1, \infty]$ trace exactly the Petz curve on $[1, 2]$. By Theorem A, every $\tilde{\vartheta}_\beta$ with $\beta \geq 1$ is therefore a point of $X(\mathcal{G}, \leq_*)$ as well. What happens below order one is discussed in Section 7.

3.5 The tower identity

The key algebraic fact is that compressing in two steps is the same as compressing in one.

Lemma 3.6 (tower identity). *Let $P \leq Q$ be projections and $s \in [-1, 1]$. Compute the outer compression below inside $\text{ran } Q$, where $K_s(\sigma, Q)$ is faithful. Then*

$$K_s(K_s(\sigma, Q), P) = K_s(\sigma, P).$$

Consequently, if $z = Z_s(\sigma, Q)$, then $Z_s(K_s(\sigma, Q)/z, P) = Z_s(\sigma, P)/z$.

Proof. For $s \neq 0$, the s -th power of $K_s(\sigma, Q)$ inside $\text{ran } Q$ is $Q\sigma^s Q$, and compressing further to $P \leq Q$ gives $P\sigma^s P$. For $s = 0$ the same computation works with logarithms. The second statement follows from homogeneity. $\square$

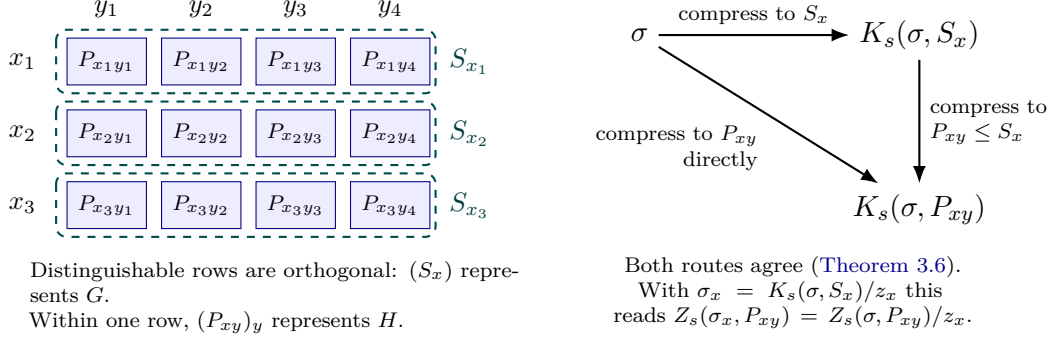


Figure 3: The proof of Theorem 4.1. Left: a representation of $G \boxtimes H$ that need not be a tensor product, organized by rows. Right: the tower identity lets each row be analysed with its own reduced center, so the weight of a cell factors into a G -part z_x and an H -part.

4 Every order up to two is a spectral point

We need four properties: normalization, additivity under $\sqcup$, multiplicativity under $\boxtimes$, and monotonicity under $\leq_*$. Normalization is immediate. On K_1 the divergence is nonnegative and vanishes at $\rho = \sigma$, so $\vartheta_\alpha(K_1) = 1$. The product is the heart of the matter, so we start there.

4.1 Products

Here is the idea (Fig. 3). Take any representation (P_{xy}) of $G \boxtimes H$ and group its projections by the first coordinate. The ranges in row x span a subspace, with projection S_x . If x and x' are distinguishable in G , then every cell of row x is distinguishable from every cell of row x' , so $S_x \perp S_{x'}$ and the rows represent G . Inside one row, the cells represent H . All that is missing is a way to split the weight of a cell into a row part and a within-row part. The tower identity does exactly this.

Theorem 4.1 (products). *For $0 \leq \alpha \leq 2$ and all graphs G, H ,*

$$\vartheta_\alpha(G \boxtimes H) = \vartheta_\alpha(G) \vartheta_\alpha(H).$$

Proof. If either graph is empty both sides vanish, so assume both are nonempty and put $s = 1 - \alpha$.

For “ $\leq$ ”, take witnesses $(\rho_x), \sigma$ for G and $(\tau_y), \omega$ for H . If (x, y) and (x', y') are distinct and non-adjacent in $G \boxtimes H$, then x, x' are distinct and non-adjacent in G or y, y' are in H . In either case $(\rho_x \otimes \tau_y)(\rho_{x'} \otimes \tau_{y'}) = 0$. Since D_α is additive under tensor products, radii add.

For “ $\geq$ ”, let (P_{xy}) represent $G \boxtimes H$, let σ be faithful, and suppose $Z_s(\sigma, P_{xy}) \geq c > 0$ for every pair (x, y) . Let S_x be the projection onto $\sum_y \text{ran } P_{xy}$, and put $z_x = Z_s(\sigma, S_x)$.

The rows represent G . If $x \neq x'$ are non-adjacent in G , every pair $(x, y), (x', y')$ is distinct and non-adjacent in $G \boxtimes H$, so $P_{xy}P_{x'y'} = 0$ and hence $S_x S_{x'} = 0$.

Each row represents H . Fix x . If $y \neq y'$ are non-adjacent in H , then (x, y) and (x, y') are non-adjacent, so $(P_{xy})_y$ represents H on $\text{ran } S_x$. Use the faithful state $\sigma_x = K_s(\sigma, S_x)/z_x$ on $\text{ran } S_x$ as center. By Theorem 3.6, $Z_s(\sigma_x, P_{xy}) = Z_s(\sigma, P_{xy})/z_x \geq c/z_x$ for every y . The projection formula (11) for H gives $c/z_x \leq 1/\vartheta_\alpha(H)$, that is, $z_x \geq c\vartheta_\alpha(H)$.

Combine. Apply (11) for G to (S_x) and σ : $c\vartheta_\alpha(H) \leq \min_x z_x \leq 1/\vartheta_\alpha(G)$. So $c \leq$

$1/(\vartheta_\alpha(G)\vartheta_\alpha(H))$ for every admissible c , and taking the supremum gives

$$\frac{1}{\vartheta_\alpha(G \boxtimes H)} \leq \frac{1}{\vartheta_\alpha(G)\vartheta_\alpha(H)}. \quad \square$$

Remark 4.2. At $\alpha = 0$ and $\alpha = 1$, [Theorem 4.1](#) recovers the multiplicativity of ϑ [[Lov79](#)] and Duan and Winter’s additivity of $C_{\min}$ [[DW16](#), Lemma 23]. At $\alpha = 2$ it says $\Sigma(G \boxtimes H) = \Sigma(G)\Sigma(H)$. Hence $S_{0,\text{NS}}(G) = \log \Sigma(G)$: the regularization in Duan and Winter’s asymptotic simulation cost is unnecessary at the graph level. Their fixed-cq-graph result [[DW16](#)] is the special case $P_{xy} = P_x \otimes Q_y$.

4.2 Disjoint unions

Two small lemmas do the work. The first rewrites a divergence after compressing the center. The second says that the weights of orthogonal blocks never add up to more than one.

Lemma 4.3 (compression identity). *Let $Q \neq 0$ be a projection, $s = 1 - \alpha \in [-1, 1]$, and $\sigma_Q = K_s(\sigma, Q)/Z_s(\sigma, Q)$. Every state ρ supported in $\text{ran } Q$ satisfies*

$$D_\alpha(\rho \parallel \sigma) = D_\alpha(\rho \parallel \sigma_Q) - \log Z_s(\sigma, Q).$$

Proof. For $\alpha \notin \{0, 1\}$, substitute $Q\sigma^s Q = Z_s^s \sigma_Q^s$ into the Petz moment. For $\alpha = 1$ use $Q(\ln \sigma)Q = \ln \sigma_Q + (\ln Z_0)Q$ on $\text{ran } Q$, and for $\alpha = 0$ use $Q\sigma Q = Z_1 \sigma_Q$. $\square$

Lemma 4.4 (orthogonal blocks). *If $Q_1, \dots, Q_m$ are mutually orthogonal projections, σ is a faithful state and $s \in [-1, 1]$, then $\sum_j Z_s(\sigma, Q_j) \leq 1$.*

The proof is a one-line application of Jensen’s inequality ([Section B.2](#)).

Theorem 4.5 (disjoint unions). *For $0 \leq \alpha \leq 2$, $\vartheta_\alpha(G \sqcup H) = \vartheta_\alpha(G) + \vartheta_\alpha(H)$.*

Proof. Assume both graphs are nonempty. For “ $\leq$ ”, put witnesses for G and H with radii at most $\log a$ and $\log b$ on orthogonal summands, and use the center $(a\sigma_G \oplus b\sigma_H)/(a+b)$. Compressing this center to the first summand gives weight $a/(a+b)$ and reduced center σ_G , so by [Theorem 4.3](#) every G -state has radius at most $\log a + \log \frac{a+b}{a} = \log(a+b)$. The same holds for H .

For “ $\geq$ ”, take a compatible family of $G \sqcup H$ with center σ and radius at most R . Let Q_G and Q_H project onto the spans of the supports of the two subfamilies. Cross pairs are non-adjacent, so $Q_G Q_H = 0$. By [Theorem 4.3](#), the G -states have radius at most $R + \log Z_s(\sigma, Q_G)$ around σ_{Q_G} , so $\vartheta_\alpha(G) \leq 2^R Z_s(\sigma, Q_G)$. Likewise $\vartheta_\alpha(H) \leq 2^R Z_s(\sigma, Q_H)$. Add the two bounds and apply [Theorem 4.4](#). $\square$

In particular $\vartheta_\alpha(\overline{K_n}) = n$.

4.3 Entanglement-assisted monotonicity

The proof follows the protocol described after (10). Bob ends up holding his share of the entanglement together with the channel output. If τ_h is the output for h , his joint state given g is $\sum_h \omega_g^h \otimes \tau_h$. These joint states form a new channel for G , and the only question is whether its radius can be larger than the old one. It cannot. The joint state is a flagged mixture with the flag discarded, and discarding the flag is data processing.

Lemma 4.6 (flagged mixtures). *Let $\omega_h \geq 0$ with $\sum_h \omega_h = \omega$ and $\text{Tr } \omega = 1$, let τ_h be states and let σ be faithful. If $D_\alpha(\tau_h \| \sigma) \leq R$ whenever $\omega_h \neq 0$, then for $0 \leq \alpha \leq 2$*

$$D_\alpha\left(\sum_h \omega_h \otimes \tau_h \parallel \omega \otimes \sigma\right) \leq R.$$

Proof. Add a classical flag and compare $\Omega = \sum_h |h\rangle\langle h| \otimes \omega_h \otimes \tau_h$ with $\Omega' = \sum_h |h\rangle\langle h| \otimes \omega_h \otimes \sigma$; the support of Ω lies in that of Ω' . The Petz moment splits over the flag, and the ω_h factors cancel:

$$\text{Tr } \Omega^\alpha \Omega'^{1-\alpha} = \sum_h w_h \text{Tr } \tau_h^\alpha \sigma^{1-\alpha}, \quad w_h = \text{Tr } \omega_h, \quad \sum_h w_h = 1.$$

For $0 < \alpha < 1$ each term satisfies $\text{Tr } \tau_h^\alpha \sigma^{1-\alpha} \geq 2^{(\alpha-1)R}$, so the average does too and $D_\alpha(\Omega \| \Omega') \leq R$. For $1 < \alpha \leq 2$ all inequalities reverse and the conclusion is the same. At $\alpha = 1$ the divergence is the w -average of the $D(\tau_h \| \sigma)$, and at $\alpha = 0$ we get $\text{Tr } \Pi_\Omega \Omega' = \sum_h w_h \text{Tr } \Pi_{\tau_h} \sigma \geq 2^{-R}$. Tracing out the flag maps $\Omega \mapsto \sum_h \omega_h \otimes \tau_h$ and $\Omega' \mapsto \omega \otimes \sigma$, and Petz divergences of order $\alpha \in [0, 2]$ do not increase under partial traces [Pet86, Tom16]. $\square$

Theorem 4.7 (entanglement-assisted monotonicity). *If $G \leq_* H$, then $\vartheta_\alpha(G) \leq \vartheta_\alpha(H)$ for every $\alpha \in [0, 2]$.*

Proof. If G is empty there is nothing to prove, and if G is nonempty then (9) forces H to be nonempty. Take ω, ω_g^h as in (9)–(10), restricted to $\text{supp } \omega$, and a compatible family (τ_h) for H with faithful center σ and radius at most R . Define

$$\tau'_g = \sum_h \omega_g^h \otimes \tau_h, \quad \sigma' = \omega \otimes \sigma.$$

Each τ'_g has trace $\text{Tr } \omega = 1$. If $g \neq g'$ are non-adjacent, every term of $\tau'_g \tau'_{g'} = \sum_{h,h'} \omega_g^h \omega_{g'}^{h'} \otimes \tau_h \tau_{h'}$ vanishes. When $h \simeq h'$ this is (10). Otherwise h, h' are distinct and non-adjacent in H , and then $\tau_h \tau_{h'} = 0$. So (τ'_g) is compatible with G , and Theorem 4.6 bounds its radius around σ' by R . $\square$

Proof of Theorem A. Normalization was noted at the start of the section. Additivity is Theorem 4.5, multiplicativity is Theorem 4.1, and monotonicity is Theorem 4.7. $\square$

Remark 4.8 (orders above two). The only place the restriction $\alpha \leq 2$ enters is data processing in Theorem 4.6. Petz data processing fails above order two, and the inequality of Theorem 4.6 itself fails too. The verifier accompanying this paper contains a finite-dimensional order-three example that violates it by more than half a bit. So this argument says nothing about $\alpha > 2$.

5 The pentagon

Throughout this section C_5 is labelled by $\mathbb{Z}_5$ with $i \sim i \pm 1$, and a representation satisfies $P_i P_{i+2} = 0$ for every i . We begin at order one half, where everything is exact, and then work down to order zero.

5.1 Order one half and above

Lemma 5.1 (Hilbert–Schmidt inequality). *For every representation of C_5 and every operator X ,*

$$\sum_i \|P_i X P_i\|_2^2 \leq 2 \|X\|_2^2.$$

Proof. Put $T = \sum_i P_i$. The terms $P_i P_{i\pm 2} P_i$ vanish, and P_{i-1} and P_{i+1} are orthogonal to each other, so

$$P_i T P_i = P_i + P_i(P_{i-1} + P_{i+1})P_i \leq 2P_i. \quad (13)$$

Consider the map $\mathcal{A} : (X_i)_i \mapsto \sum_i X_i$ from $\bigoplus_i P_i \mathcal{B}_2 P_i$ to the Hilbert–Schmidt class $\mathcal{B}_2$. For $X_i = P_i X P_i$, cyclicity gives $\text{Tr } X_i^* X_j = \text{Tr}((P_j X_i)^*(X_j P_i))$, so

$$|\text{Tr } X_i^* X_j| \leq \frac{1}{2}(\|P_j X_i\|_2^2 + \|X_j P_i\|_2^2).$$

Summing over i and j and using (13),

$$\left\| \sum_i X_i \right\|_2^2 \leq \frac{1}{2} \sum_i [\text{Tr}(X_i^* T X_i) + \text{Tr}(X_i T X_i^*)] \leq 2 \sum_i \|X_i\|_2^2.$$

So $\|\mathcal{A}\|^2 \leq 2$. The adjoint of $\mathcal{A}$ is $X \mapsto (P_i X P_i)_i$, and it has the same norm. $\square$

Theorem 5.2. $\vartheta_\alpha(C_5) = \frac{5}{2}$ for every $\alpha \in [\frac{1}{2}, 2]$.

Proof. At $s = \frac{1}{2}$ the weight is $Z_{1/2}(\sigma, P) = \|P\sigma^{1/2}P\|_2^2$. **Theorem 5.1** with $X = \sigma^{1/2}$ gives $\sum_i Z_{1/2}(\sigma, P_i) \leq 2 \text{Tr } \sigma = 2$. So the minimum over i is at most $\frac{2}{5}$, and (11) gives $\vartheta_{1/2}(C_5) \geq \frac{5}{2}$. By monotonicity in α the same lower bound holds on $[\frac{1}{2}, 2]$. For the upper bound, $\alpha^*(C_5) = \frac{5}{2}$ and **Theorem 3.3**(iii) applies. Explicitly, the flat witness is $\rho_i = \frac{1}{2}(|i\rangle\langle i| + |i+1\rangle\langle i+1|)$ on $\mathbb{C}^5$ with $\sigma = I/5$, and it has radius $\log \frac{5}{2}$ at every order. $\square$

5.2 Below order one half

Order zero is governed by a trace-norm version of **Theorem 5.1**, with the constant $\sqrt{5}$ in place of 2. Interpolating between the two gives everything in between.

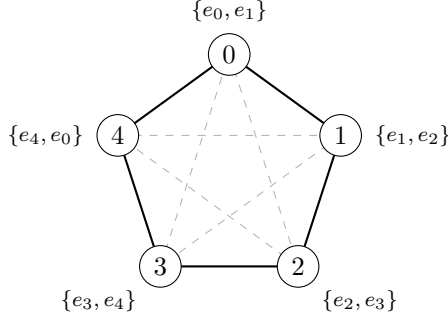
Lemma 5.3 (trace-norm inequality). *For every representation of C_5 and every operator X ,* $\sum_i \|P_i X P_i\|_1 \leq \sqrt{5} \|X\|_1$.

Proof. If (P_i) represents G and v is a vector, pick unit vectors $u_i \in \text{ran } P_i$ parallel to $P_i v$ (arbitrary if $P_i v = 0$). They form an orthonormal representation of G , and $\sum_i \|P_i v\|^2 = \sum_i |\langle u_i, v \rangle|^2 \leq \vartheta(\overline{G}) \|v\|^2$ by Lovász’s theorem [Lov79]. The pentagon is self-complementary, so the constant is $\sqrt{5}$. For $X = ab^*$ we have $\|P_i ab^* P_i\|_1 = \|P_i a\| \|P_i b\|$, and Cauchy–Schwarz proves the claim for rank one. A singular value decomposition and the triangle inequality do the rest. $\square$

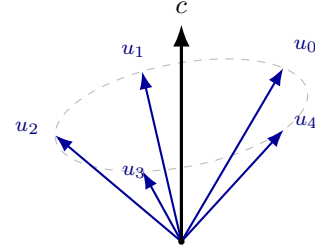
Lemma 5.4 (interpolation). *For $1 \leq p \leq 2$, every representation of C_5 and every operator X ,*

$$\sum_i \|P_i X P_i\|_p^p \leq B(p) \|X\|_p^p, \quad B(p) = 5^{(2-p)/2} 2^{p-1}.$$

Proof. The map $X \mapsto (P_i X P_i)_i$ has norm at most $\sqrt{5}$ from S_1 to $\ell_1(S_1)$ by **Theorem 5.3**, and at most $\sqrt{2}$ from S_2 to $\ell_2(S_2)$ by **Theorem 5.1**. Complex interpolation of Schatten classes and their ℓ_p -sums [BL76, Bha97] bounds its norm from S_p to $\ell_p(S_p)$ by $5^{(2-p)/(2p)} 2^{(p-1)/p}$. Raise this to the power p . $\square$



(a) flat witness, $\alpha \geq \frac{1}{2}$



(b) Lovász's umbrella, $\alpha = 0$

Figure 4: Two optimal geometries for the pentagon. Solid lines are edges and dashed chords are the non-edges, which must be orthogonal. (a) Vertex i gets the uniform state on $\{e_i, e_{i+1}\}$ and the center is $I/5$. Non-adjacent supports are disjoint, and the radius is $\log \frac{5}{2}$ at every order. (b) Unit vectors u_i on a cone around the handle c with $|\langle c, u_i \rangle|^2 = 1/\sqrt{5}$ and $u_i \perp u_{i+2}$. This attains $\vartheta(C_5) = \sqrt{5}$ at order zero, and its radius grows like $5^{1/(2-2\alpha)}$. Neither geometry is optimal for $0 < \alpha < \frac{1}{2}$.

Theorem 5.5. For $0 \leq \alpha \leq \frac{1}{2}$ and $p = 1/(1 - \alpha)$, the bounds (4) hold.

Proof. For the lower bound put $s = 1/p$ and $X = \sigma^s$ in Theorem 5.4. Then $\|X\|_p^p = \text{Tr } \sigma = 1$ and $\|P_i \sigma^s P_i\|_p^p = Z_s(\sigma, P_i)$, so $\sum_i Z_s(\sigma, P_i) \leq B(p)$. The minimum is at most the average, so $\vartheta_\alpha(C_5) \geq 5/B(p) = \sqrt{5}(\sqrt{5}/2)^{p-1}$.

The bound $\frac{5}{2}$ is Theorem 5.2 combined with monotonicity. For the other upper bound, take the umbrella of Fig. 4(b): unit vectors $u_i = (\cos \gamma, \sin \gamma \cos \frac{2\pi i}{5}, \sin \gamma \sin \frac{2\pi i}{5})$ with $\cos^2 \gamma = 1/\sqrt{5}$, which satisfy $u_i \perp u_{i+2}$. For a pure state, $\text{Tr}[u \langle u | \sigma^{1-\alpha} | u \rangle] = \langle u | \sigma^{1-\alpha} | u \rangle$. As faithful centers approach $|c \rangle \langle c|$ with $c = (1, 0, 0)$, this tends to $|\langle c, u_i \rangle|^2 = 5^{-1/2}$ because $1 - \alpha > 0$. The radius therefore tends to $\frac{1}{1-\alpha} \log \sqrt{5} = \log 5^{p/2}$. $\square$

5.3 A continuum of spectral points

Proof of Theorem B. The values on $[\frac{1}{2}, 2]$ are Theorem 5.2, and the bounds on $[0, \frac{1}{2}]$ are Theorem 5.5. By Theorems 3.3 and 3.4, $\alpha \mapsto \vartheta_\alpha(C_5)$ is continuous and nondecreasing on $[0, \frac{1}{2}]$. It runs from $\vartheta(C_5) = \sqrt{5}$ to $\frac{5}{2}$, so by the intermediate value theorem its image is $[\sqrt{5}, \frac{5}{2}]$. Parameters with different values on C_5 are different functions, and each of them lies in $X(\mathcal{G}, \leq_*)$ by Theorem A. $\square$

The lower bound in (4) exceeds $\sqrt{5}$ as soon as $\alpha > 0$. So no ϑ_α with $\alpha > 0$ equals ϑ .

Remark 5.6 (an explicit ladder). Readers who want explicit separations without the intermediate value theorem can use $\alpha_k = 1/(8^{k+1} + 1)$, $k \geq 0$. Then $p_k - 1 = 8^{-(k+1)}$, and the bounds (4) give $\vartheta_{\alpha_{k+1}}(C_5) \leq \sqrt{5} \cdot 5^{(p_k-1)/16} < \sqrt{5}(\sqrt{5}/2)^{p_k-1} \leq \vartheta_{\alpha_k}(C_5)$. The strict inequality is $5^{1/16} < \sqrt{5}/2$, that is, $2^{16} = 65536 < 78125 = 5^7$. Together with $\vartheta_{\alpha_0}(C_5) \leq 5^{9/16} < \frac{5}{2}$, this shows that $\vartheta, \vartheta_{\alpha_0}, \vartheta_{\alpha_1}, \dots, \vartheta_{1/2}$ are pairwise distinct.

5.4 The Duan–Winter chain

Duan and Winter prove the chain (1) [DW16, Eq. (54)]. They show that its last link can be strict, and they conjecture that on the pentagon every term except the first equals $\log \frac{5}{2}$. They also ask

whether the two middle links can ever be strict. We can now answer all of this.

Corollary 5.7 (the pentagon conjecture). $\log \sqrt{5} < C_{\min}(C_5) = S_{0,\text{NS}}(C_5) = \log \Sigma(C_5) = \log \alpha^*(C_5) = \log \frac{5}{2}$.

Proof. Theorems 3.3 and 5.2 give $C_{\min}(C_5) = \log \Sigma(C_5) = \log \frac{5}{2}$, and (1) squeezes $S_{0,\text{NS}}(C_5)$ between them. The fractional packing number of C_5 is $\frac{5}{2}$ and $\vartheta(C_5) = \sqrt{5}$ [Lov79]. $\square$

This proof uses only Theorem 5.1, the projection formula and monotonicity in α . The product theorem is not needed for the pentagon, but it is needed for the general statement.

Corollary 5.8 (the middle links). *For every graph G , $S_{0,\text{NS}}(G) = \log \Sigma(G)$. There is a graph with $C_{\min}(G) < S_{0,\text{NS}}(G)$.*

Proof. The first statement is Theorem 4.2. The second is Theorem C, since $C_{\min}(G_{32}) < \log \Sigma(G_{32}) = S_{0,\text{NS}}(G_{32})$. $\square$

Duan and Winter also ask whether Σ can be strictly submultiplicative [DW16]. For confusability graphs, with the optimization over compatible channels built in, Theorem 4.1 says that it cannot. Their question for general non-commutative bipartite graphs is not addressed here.

6 Order one versus order two

6.1 Where to look

Theorem 3.2 tells us where *not* to look. If some flat witness is optimal at order one, then orders one and two agree. This is what happens on the pentagon, and on every graph whose best order-one channel is flat. To separate the endpoints we need a graph whose good channels are genuinely quantum. Contextuality scenarios are a natural source, since they come with rigid noncommutative representations. We take the most familiar one, the Mermin–Peres magic square [Mer90, Per90], and then add weights by cloning vertices. This section works directly from the definitions (7) and (8) and does not use Theorem 4.1.

6.2 The graph

Index the nine cells of a 3×3 array by $(i, j) \in \{0, 1, 2\}^2$, and fill them with the two-qubit observables of Fig. 5. The six contexts are the three rows and the three columns. A vertex of the *seed graph* F is a context together with signs for its three cells: signs with product $+1$ on rows and on the first two columns, and with product -1 on the last column. Two distinct vertices are non-adjacent if they belong to the same context, or if they give opposite signs to a shared cell. That makes 24 vertices, each non-adjacent to exactly nine others (216 ordered non-adjacent pairs).

For a context whose first two observables are A and B , the vertex with first two signs (s, t) is represented by the joint eigenprojection

$$Q_{A,B;s,t} = \frac{1}{4}(I + sA)(I + tB),$$

a rank-one projection on $\mathbb{C}^2 \otimes \mathbb{C}^2$. These projections represent F . In fact $Q_v Q_w = 0$ holds exactly for the non-adjacent pairs, with no accidental orthogonality. Let vertex 0 be the first-row vertex with signs $(+, +, +)$, so $Q_0 = |00\rangle\langle 00|$, and sort the vertices by overlap $j_v = 4 \text{Tr } Q_v Q_0 \in \{0, 1, 2, 4\}$. The four classes have sizes 9, 8, 6, 1. Class 0 consists of the nine non-neighbours of vertex 0, and class 4 is vertex 0 itself.

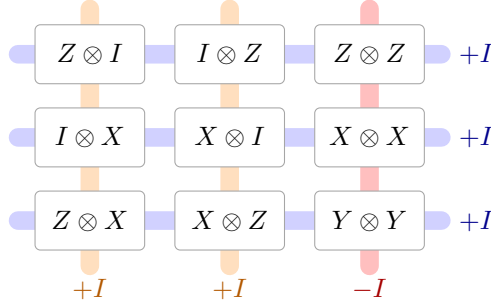


Figure 5: The Mermin–Peres magic square. Each row and each column is a *context*: its three observables commute, and their product is the operator shown at its end. A vertex of the seed graph F is a context together with a joint eigenvalue pattern (three signs with the prescribed product). That gives four vertices per context and 24 in all.

Weights enter through cloning. For a profile $k = (k_0, k_1, k_2, k_4)$ of nonnegative integers, the *clone graph* $F[k]$ replaces every class- j vertex by k_j pairwise non-adjacent copies (deleting it if $k_j = 0$). Each copy inherits its parent’s adjacencies to all other vertices. Our main example is

$$G_{32} = F[(2, 1, 1, 0)], \quad |V(G_{32})| = 2 \cdot 9 + 8 + 6 = 32.$$

6.3 Two weights and a bridge

Two of the weights of [Section 3](#) matter here. The Gibbs weight $w_\sigma(P) = Z_0(\sigma, P)$ belongs to order one, and the harmonic weight $h_\sigma(P) = Z_{-1}(\sigma, P) = \text{Tr } \sigma_{/P}$ belongs to order two. For a profile k , let $t_W(F[k])$ be the supremum of $\min_{v: k_v > 0} w_\sigma(P_v)/k_v$ over representations (P_v) of the seed F and faithful states σ . Define $t_H(F[k])$ in the same way with h_σ . Here k_v means k_{j_v} , and deleted vertices may be represented by $P_v = 0$.

Proposition 6.1 (bridge). *For every profile k ,*

$$2^{C_{\min}(F[k])} \leq \frac{1}{t_W(F[k])} \quad \text{and} \quad \Sigma(F[k]) \geq \frac{1}{t_H(F[k])}.$$

Proof. Upper bound on $C_{\min}$. Start from a representation (P_v) of F and a state σ . Let $A = \prod_{v: k_v > 0} \{1, \dots, k_v\}$, work on $\bigoplus_{f \in A} \mathcal{H}$, and put

$$\sigma' = \frac{1}{|A|} \bigoplus_{f \in A} \sigma, \quad P'_{v,a} = \bigoplus_{f \in A: f(v)=a} P_v$$

(zero in the other summands). Copies of one vertex live in disjoint sets of summands, so they are orthogonal, and non-adjacent seed vertices stay orthogonal. So $(P'_{v,a})$ represents $F[k]$. Functional calculus on the direct sum gives $w_{\sigma'}(P'_{v,a}) = w_\sigma(P_v)/k_v$. Now apply the projection formula (11) at $\alpha = 1$ together with $\vartheta_1 = 2^{C_{\min}}$.

Lower bound on Σ . Take a compatible family $(\rho_{v,a})$ for $F[k]$ and $T \geq \rho_{v,a}$, put $t = \text{Tr } T$ and $\sigma = T/t$ (regularized if needed), and let $P_{v,a}$ be the supports. As in [Theorem 3.3](#), $h_\sigma(P_{v,a}) \geq 1/t$. Copies of one vertex are non-adjacent, so $P_v = \sum_a P_{v,a}$ is a projection, and (P_v) (with $P_v = 0$ for deleted v) represents F . The harmonic weight is superadditive over orthogonal blocks: for a positive definite block matrix M , each diagonal block of M^{-1} dominates the inverse of the corresponding block of M . Applied to $M = P_v \sigma^{-1} P_v$ this gives $h_\sigma(P_v) \geq \sum_a h_\sigma(P_{v,a}) \geq k_v/t$. Hence $t_H(F[k]) \geq 1/t$. $\square$

graph	profile (k_0, k_1, k_2, k_4)	vertices	$2^{C_{\min}} \leq$	$\Sigma \geq$	gap (bits)
G_{32}	(2, 1, 1, 0)	32	6.186983...	6.194232...	0.001689
G_{44}	(1, 2, 3, 1)	44	8.949075...	8.962394...	0.002146
G_{72}	(4, 3, 2, 0)	72	105/8	13.191800...	0.007324

Table 1: Certified bounds for three clone graphs. Upper bounds on $2^{C_{\min}}$ come from explicit rational Gibbs witnesses: a direct sum of the magic-square representation and one-dimensional clique representations. Lower bounds on Σ come from rational dual certificates for the moment relaxation.

The upper bound on $C_{\min}$ needs one good witness. The lower bound on Σ needs an upper bound on t_H valid for *every* representation, in every dimension. That is the job of the next subsection.

6.4 Harmonic weights as moments

Let (P_v) represent F on $\mathcal{H}$ and let σ be faithful. Put $\Omega = \text{vec}(\sqrt{\sigma}) \in \mathcal{H} \otimes \mathcal{H}$, let F_v be the orthogonal projection onto $\sigma^{-1/2} \text{ran } P_v$, and let $E_v = F_v^\top$.

Lemma 6.2 (harmonic weight as a moment). *For every v ,*

$$(P_v \otimes E_v)\Omega = (I \otimes E_v)\Omega \quad \text{and} \quad \langle \Omega, (I \otimes E_v)\Omega \rangle = h_\sigma(P_v).$$

Proof. With $(A \otimes B) \text{vec}(M) = \text{vec}(AMB^\top)$, the first identity says $P_v \sqrt{\sigma} F_v = \sqrt{\sigma} F_v$. This holds because $\sqrt{\sigma} F_v$ has range $\text{ran } P_v$. For the second, $\sqrt{\sigma} F_v \sqrt{\sigma}$ lies below σ and has range in $\text{ran } P_v$. If $0 \leq A \leq \sigma$ has range in $\text{ran } P_v$, then $\sigma^{-1/2} A \sigma^{-1/2}$ is a contraction supported in $\text{ran } F_v$, hence at most F_v . So $\sqrt{\sigma} F_v \sqrt{\sigma} = \sigma|_{P_v}$, and its trace is the moment. $\square$

So harmonic weights are expectations in the vector Ω of two commuting families of projections, $P_v \otimes I$ and $I \otimes E_v$. The first family obeys the orthogonality relations of F . The second obeys no relations of its own, but it is tied to the first by the identity of [Theorem 6.2](#). This is exactly the setting of noncommutative moment relaxations [[NPA08](#), [Mgh26](#)]. We form the Gram matrix of the 409 vectors

$$\Omega, \quad (P_a \otimes I)\Omega, \quad (I \otimes E_b)\Omega, \quad (P_a \otimes E_b)\Omega \quad (a = b \text{ or } a \sim b).$$

The pairs with $a \not\sim b$ are left out because those vectors vanish: $(P_a \otimes E_b)\Omega = (P_a P_b \otimes E_b)\Omega = 0$. The Gram matrix is positive semidefinite, its corner entry is $\text{Tr } \sigma = 1$, and the entry $\langle \Omega, (I \otimes E_v)\Omega \rangle$ is $h_\sigma(P_v)$. Many entries coincide or vanish in every model. We record these coincidences in a table of 1009 moment keys, using four rewriting rules together with symmetry ([Section A](#)). Maximizing t subject to positivity, the corner normalization and $h_v \geq t k_v$ gives an upper bound on $t_H(F[k])$ in every dimension. A rational dual solution then certifies a rational upper bound.

6.5 Certified values

[Table 1](#) lists the certified bounds. For G_{32} the exact rational values are

$$t_W(G_{32}) \geq \frac{5225365592240894}{32329248967670751} > \frac{15147154828811}{93824992214030} \geq t_H(G_{32}). \quad (14)$$

For G_{44} and G_{72} the certified harmonic bounds are $t_H \leq 31406288103814/281475531618259$ and $t_H \leq 21337116511217/281474976708840$.

Proof of Theorem C. Combine (14) with Theorem 6.1. The reciprocals of the two rationals are 6.186983... and 6.194232.... $\square$

7 Open problems

Is ϑ the least point? Every ϑ_α dominates ϑ , so this paper leaves the conjecture $\Theta_* = \vartheta$ where it was. A proof that ϑ is the minimum of $X(\mathcal{G}, \leq_*)$ would settle it. So would a single spectral point below ϑ on some graph, in the opposite direction.

Above order two. Theorem 4.8 shows that our monotonicity argument breaks down above order two. Is some suitably defined ϑ_α with $\alpha > 2$ still in $X(\mathcal{G}, \leq_*)$? Flat witnesses show that any such extension is at most α^* . A natural guess for the $\alpha \rightarrow \infty$ endpoint is the complement of the projective rank. Flat representations give the corresponding upper bound, but we are missing a dimension-free argument for the other direction.

Other divergences. The proof of Theorem A uses three features of the Petz family. These are a closed form for the supported minimum with the tower property, the block structure used for unions and flagged mixtures, and data processing. For sandwiched divergences of order $\beta \geq 1$ the question has a complete answer: by Theorem 3.5 they reproduce the Petz curve on $[1, 2]$, from $2^{C_{\min}}$ at $\beta = 1$ to Σ at $\beta = \infty$. Sandwiched divergences also satisfy data processing for $\frac{1}{2} \leq \beta < 1$ [Tom16], but there we have no closed form for the supported minimum. Do these orders give spectral points, and do they differ from the ϑ_α with $\alpha < 1$?

Convexity. In the ordinary spectrum, Vrana’s probabilistic refinement makes the spectrum log-convex [Vra21]. If the same held for $\leq_*$, interpolating between ϑ and Σ would produce another continuum. At least part of the Rényi family is not of this form. Along such interpolations, vertex-transitive graphs are evaluated log-affinely [Vra22]. A point with value $\frac{5}{2}$ on C_5 would then carry no weight on ϑ , so it would be Σ itself. But $\vartheta_1(C_5) = \frac{5}{2}$ and $\vartheta_1 \neq \Sigma$ by Theorem C. So ϑ_1 is not such an interpolant, and we do not know whether the ϑ_α with $\alpha < \frac{1}{2}$ are. Making any of this rigorous would need an entanglement-assisted theory of probabilistic refinements, which we do not have.

The pentagon curve. What is $\vartheta_\alpha(C_5)$ for $0 < \alpha < \frac{1}{2}$? The bounds in Fig. 1 do not meet, and neither geometry in Fig. 4 is optimal there.

Small separations. How small can a graph with $C_{\min} < \log \Sigma$ be? Thirty-two vertices suffice, and we have no reason to think this is minimal. Can the rational certificate behind Theorem C be replaced by a short analytic inequality, in the spirit of Theorem 5.1? And is the first level of the moment relaxation already exact for the three graphs of Table 1?

Non-commutative graphs. Duan, Severini and Winter’s quantum Lovász number extends ϑ to non-commutative graphs [DSW13]. Does the whole family ϑ_α extend, and does the product theorem survive? A positive answer at order two would bear on Duan and Winter’s question about strict submultiplicativity of Σ for general non-commutative bipartite graphs.

Reproducibility. The certificate data and the verification scripts accompany this paper. They are needed only for the finite rational claims of [Section 6](#). Everything else is proved in the text.

AI-assisted tools. Generative AI tools were used throughout the research and preparation of this manuscript, including for mathematical exploration, proof development and critique, literature organization, computational assistance, and drafting. The authors determined which results and arguments to include and take responsibility for the final manuscript. Exact computer-assisted claims are supported, where indicated, by accompanying deterministic certificates and verification scripts.

A The certificate

Words and moments. For words $L = a_1 \cdots a_m$ and $R = b_1 \cdots b_n$ in seed vertices, write $P_L = P_{a_1} \cdots P_{a_m}$, $E_R = E_{b_1} \cdots E_{b_n}$ and

$$m(L, R) = \langle \Omega, (P_L \otimes E_R) \Omega \rangle.$$

Every entry of the Gram matrix of [Section 6](#) is such a moment. The following rules hold in every model.

- (R1) Equal adjacent letters in L or in R merge, since $P_a^2 = P_a$ and $E_b^2 = E_b$.
- (R2) If L contains two adjacent letters $a \neq b$ that are non-adjacent in F , then $m(L, R) = 0$, since $P_a P_b = 0$.
- (R3) If R ends in y , then $m(L, R) = m(Ly, R)$, since $(I \otimes E_y) \Omega = (P_y \otimes E_y) \Omega$.
- (R4) If R begins with y , then $m(L, R) = m(yL, R)$, by the adjoint of the same identity.

Reversing both words conjugates the moment, since $(P_L \otimes E_R)^* = P_{\bar{L}} \otimes E_{\bar{R}}$. Let $\mathcal{A}_0$ be the 48 automorphisms of F that fix vertex 0 (equivalently, that preserve the overlap classes). A relabelled representation is again a representation, and the constraints $h_v \geq tk_v$ are invariant because k is constant on classes. So we may average the Gram matrix over $\mathcal{A}_0$ and then take real parts. Both operations preserve positivity, and together they identify symmetry-related moments and reversed pairs. The verifier builds the key table from these rules only. So every identification in the table holds for the Gram matrix of every representation, in every dimension. The result has 1009 keys.

Deleted vertices. In G_{32} and G_{72} the class-4 vertex is deleted. It stays in the seed and is represented by $P_0 = 0$, and hence $E_0 = 0$. All four rules remain valid for zero projections, no radial constraint is imposed on deleted vertices, and the relaxation needs no change.

The relaxation and its dual. Let $\Gamma(y)$ be the real symmetric matrix with one variable y_κ per key, and let κ_v be the key of $m(\emptyset, v) = h_\sigma(P_v)$. The primal problem is to maximize t subject to

$$\Gamma(y) \succeq 0, \quad y_1 = 1, \quad y_{\kappa_v} \geq tk_v \quad (k_v > 0).$$

A dual certificate consists of $Z \succeq 0$, multipliers $\mu_v \geq 0$ and a number λ such that, for every key κ ,

$$\sum_{(i,j): \text{key}(i,j)=\kappa} Z_{ij} + \sum_{v: \kappa_v=\kappa} \mu_v = \lambda [\kappa = 1]. \quad (15)$$

Pairing with $\Gamma(y)$ gives $0 \leq \langle Z, \Gamma(y) \rangle = \lambda - \sum_v \mu_v y_{\kappa_v}$, and therefore

$$t_H(F[k]) \leq \frac{\lambda}{\sum_v \mu_v k_v}.$$

Exact positivity. The file `certificates/small_graph_certificates.json` contains, for each graph, the rational Gibbs witness, the multipliers μ , and the nonzero orbits of Z . The verifier rebuilds the seed, its 216 ordered non-adjacent pairs, the two-qubit representation, the overlap classes, the 48 automorphisms, the 409 monomials and the key table. It checks every equation (15) and $\mu \geq 0$ in rational arithmetic, and it recomputes the Gibbs witness exactly. To prove $Z \succeq 0$, it clears denominators, $DZ \in \mathbb{Z}^{409 \times 409}$, rounds a floating-point Cholesky factor at scale 2^{40} to an integer matrix L , and checks in exact integer arithmetic that

$$R = 2^{80}DZ - DLL^\top$$

is symmetric and diagonally dominant with nonnegative diagonal. Then $R \succeq 0$ and $Z = LL^\top/2^{80} + R/(2^{80}D) \succeq 0$. Only the integer check counts. Whether the floating-point factorization succeeded plays no role.

Negative controls. The verifier runs two tampered certificates, and both must be rejected. One breaks an affine equation. The other changes only entries attached to moments that vanish identically, so every affine equation still holds but exact positivity fails.

How the certificate was found. We followed the recovery procedure of [Mgh26]. A numerical SDP solution is symmetrized, rounded to dyadic rationals, and projected exactly onto the affine constraints by a minimum-norm correction, and positivity is then certified as above. None of this matters for correctness, which rests only on the exact checks.

B Deferred proofs

B.1 Continuity in the order

Proof of Theorem 3.4. Write $f(\alpha) = \log \vartheta_\alpha(G)$.

Right continuity at $\alpha_0 \in [0, 1)$. Pick a compatible family and a faithful center whose order- α_0 radius is at most $f(\alpha_0) + \varepsilon$. For fixed states with a faithful center, $\alpha \mapsto D_\alpha$ is continuous on $[0, 1)$, including from the right at 0, because $\rho^\alpha \rightarrow \Pi_\rho$. So $\limsup_{\alpha \downarrow \alpha_0} f(\alpha) \leq f(\alpha_0) + \varepsilon$, and monotonicity gives the reverse inequality.

Left continuity at $\alpha \in (0, 1)$. Let $0 < \beta < \alpha$ and $\psi(t) = \ln \text{Tr } \rho^t \sigma^{1-t}$. By Hölder's inequality ψ is convex, and $\psi(1) = 0 \geq \psi(0)$. So $\psi(\beta) \leq (1 - \frac{\beta}{\alpha})\psi(0) + \frac{\beta}{\alpha}\psi(\alpha) \leq \frac{\beta}{\alpha}\psi(\alpha)$. Since both orders lie below one, this rearranges to

$$D_\alpha(\rho \parallel \sigma) \leq \frac{\alpha(1 - \beta)}{\beta(1 - \alpha)} D_\beta(\rho \parallel \sigma).$$

The constant does not depend on the states or the dimension. Applying it to near-optimal witnesses at order β gives $f(\beta) \leq f(\alpha) \leq \frac{\alpha(1 - \beta)}{\beta(1 - \alpha)} f(\beta)$, and the constant tends to 1 as $\beta \uparrow \alpha$. $\square$

B.2 Orthogonal blocks

Proof of Theorem 4.4. For $s \neq 0$ let $A = \sigma^s$ and $f(t) = t^{1/s}$. For $s = 0$ let $A = \ln \sigma$ and $f = \exp$. In each case f is convex on the spectrum of A and $f(A) = \sigma$. Diagonalize $Q_j A Q_j$ on $\text{ran } Q_j$ in an orthonormal basis $(e_{jk})_k$. Its eigenvalues are $\langle e_{jk}, A e_{jk} \rangle$, so Jensen's inequality for the spectral measure of A gives

$$Z_s(\sigma, Q_j) = \sum_k f(\langle e_{jk}, A e_{jk} \rangle) \leq \sum_k \langle e_{jk}, \sigma e_{jk} \rangle.$$

The vectors $(e_{jk})_{j,k}$ are orthonormal, so the total is at most $\text{Tr } \sigma = 1$. $\square$

B.3 Exact confusability graphs

Lemma B.1. *In (6), (7) and (8), restricting to channels whose confusability graph is exactly G does not change the infimum.*

Proof. Restricting can only increase an infimum, so we show the reverse. Take a compatible family (ρ_x) on $\mathcal{H}$ with center σ (or dominator T). Add an orthonormal vector f_e for each edge e of G . For a vertex x of degree $d_x > 0$, let Π_x project onto the f_e with $e \ni x$, and let Π_E project onto all of them. Put

$$\rho'_x = (1 - \varepsilon)\rho_x \oplus \frac{\varepsilon}{d_x}\Pi_x, \quad \sigma' = (1 - \varepsilon)\sigma \oplus \frac{\varepsilon}{|E|}\Pi_E, \quad T' = T \oplus \varepsilon\Pi_E,$$

and leave isolated vertices unchanged. Adjacent x, y now overlap on $f_{\{x,y\}}$, while non-adjacent vertices share no new vector. So the confusability graph is exactly G . Every Petz moment, relative entropy and support weight splits over the two summands, and the new summand contributes $O(\varepsilon)$. So radii converge to the original ones as $\varepsilon \rightarrow 0$. Also $T' \geq \rho'_x$ and $\text{Tr } T' \rightarrow \text{Tr } T$. The statement for $C_{\min}$ follows from the order-one radius identity. $\square$

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